

Deconstructing Out-of-Distribution Extrapolation in Molecular Deep Learning: Structural Complexity, Topological Rescue, and Conformal Triage on African Antimalarial Natural Product Scaffolds

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Abstract

Graph neural networks (GNNs) and molecular transformers frequently underperform classical fingerprint-based tree ensembles under scaffold separation, yet the structural mechanics governing this extrapolation deficit remain poorly understood. On a frozen panel of 19 836 African antimalarial natural products and synthetic derivatives—characterised by the African-Chemotype Structural Index (ACSI) and exhibiting a structural paradox of 92.6 % whole-molecule novelty yet 69.3 % conservation of privileged African ring cores—derived from the African Natural Products Database (ANPDB / AfroDb), we deconstruct representation performance across 25 fold-seed replicates per model using a distance-resolved Molecular Out-of-Distribution (MOOD) framework. Under Bemis–Murcko scaffold separation, ECFP4–RF reached mean test ROC AUC 0.830 0, statistically outperforming GIN–TFP (0.813 8), GIN–TNE (0.809 0), GIN (0.804 7) and ChemBERTa (0.786 7) after Benjamini–Hochberg correction. Nearest-training-neighbour Tanimoto decile stratification reveals that the ECFP4 advantage is strictly localized in the structural extrapolation tail (deciles D1–D4, NN-similarity < 0.40, $\Delta \in 0.016 - -0.027$), whereas deep models achieve near-parity in mid-range interpolation (D5–D8, $\Delta \in 0.001 - -0.007$). Fusing multi-scale persistent-homology images into a graph isomorphism network (GIN–TFP) recovers up to 36 % of the extrapolation gap, overcoming the 1-WL expressivity bottleneck on complex cyclic natural product scaffolds ($Fsp^3 \geq 0.45$, ring count ≥ 4). To exploit these dynamics prospectively, we formulate a Distance-Aware Conformal Triage Filter combining Tanimoto distance with expected calibration error. This decision rule dynamically routes low-similarity OOD candidates to ECFP4–RF and mid-range candidates to GIN–TFP, increasing virtual screening precision by 14.2 % over single-model baselines. Finally, we demonstrate that fold-independent weight restoration in transformers is essential to eliminate cross-fold information leakage. This work establishes a mechanistic and operational foundation for molecular representation deployment under chemical distribution shift.

Keywords: graph neural networks; molecular fingerprints; African natural products; ANPDB; African-Chemotype Structural Index; 1-Weisfeiler-Lehman expressivity; scaffold split; persistent homology; conformal triage; out-of-distribution generalisation

Highlights:

- **Distance-Resolved Mechanics:** Nearest-neighbour decile stratification proves ECFP4 superiority is strictly confined to the OOD extrapolation tail (similarity < 0.40), achieving statistical parity in mid-range interpolation.
- **Topological Rescue Mechanism:** Persistent-homology graph fusion (GIN–TFP) recovers up to 36% of the extrapolation deficit by mitigating the 1-WL graph expressivity bottleneck on 3D polycyclic African natural product scaffolds ($Fsp^3 \geq 0.45$).
- **Conformal Triage Filter:** A distance- and calibration-aware decision rule dynamically routes candidate molecules, yielding a +14.2% precision boost in OOD virtual screening.

- **Leakage-Free Protocol:** Fold-independent weight restoration eliminates cross-fold information leakage in molecular sequence transformers during cross-validation.

1 Introduction

Malaria continues to impose a severe public health burden in sub-Saharan Africa, with approximately 249 million cases and 608 000 deaths recorded in 2023 [1]. The emergence of artemisinin-resistant *Plasmodium falciparum* strains across Southeast Asia and Africa [2, 3] underscores the urgent mandate, endorsed by the World Health Organization, to discover novel chemical scaffolds that remain active against drug-resistant parasites. African natural products derived from medicinal flora such as *Cryptolepis sanguinolenta*, *Enantia chlorantha*, and *Nauclea latifolia* provide a rich, stereochemically complex reservoir of such privileged scaffolds [4, 5]. The screening panel examined here originates from a campaign encompassing 396 African natural products from the African Natural Products Database (ANPDB / AfroDb) systematically expanded into a library of 19 836 antimalarial candidates, establishing direct continuity with our companion studies on polypharmacological resistance-reversal screening [6] and quantum/topological molecular representations [7]. These natural product scaffolds exhibit an inherent structural novelty paradox: while 92.6 % of the expanded molecules represent whole-molecule chemical novelty (Tanimoto < 0.40 relative to standard commercial synthetic screening libraries), they retain 69.3 % of privileged African natural product ring cores, encompassing indoloquinoline alkaloids (such as cryptolepine), protoberberine alkaloids, monoterpene indole alkaloids, and quassinoids/prenylated flavonoids. Characterised by elevated fractions of sp^3 -hybridized carbons ($F_{sp^3} \geq 0.45$), spatial stereocenters, macrocyclic cages, and bridged polycyclic frameworks, these scaffolds present a formidable challenge to molecular machine learning architectures. Furthermore, to ensure computational sovereignty for academic institutions in malaria-endemic regions, our centroid-based clustering framework achieves a 99.3 % compute overhead reduction (evaluating 1 936 representative centroids versus 263 000 full-matrix pairwise calculations), enabling high-throughput virtual screening workflows on high-performance computing (HPC) infrastructure used without dependence on external commercial cloud infrastructure. While computational virtual screening promises to accelerate hit identification, the choice of molecular representation fundamentally governs both predictive accuracy and out-of-distribution generalisation.

A scale-centric paradigm in molecular machine learning posits that pretrained foundation models and deep sequence transformers should supersede compact, task-trained cheminformatics baselines [8, 9]. Parallel developments suggest that quantum-inspired and topological representations offer further accuracy gains [7]. However, rigorous empirical benchmarks present a more tempered perspective. Across 156 task-metric evaluations, Guo and Ding [10] established that classical tree ensembles on fingerprints won 47.4 % of tasks, outperforming pretrained sequence models (28.8 %) and graph neural networks (21.8 %). An independent survey of 25 pretrained embeddings across 25 datasets concluded that almost no neural embedding statistically improved upon an ECFP baseline [11]. For natural products specifically, a 20-fingerprint benchmark across $> 100\,000$ compounds confirmed that no single encoding dominates and ECFP is frequently matched or exceeded [12].

Under chemical-distribution shift, the central research question is not merely whether deep architectures outperform classical fingerprints, but how their fundamental inductive biases dictate structural extrapolation across chemical space. Specifically: *How do structural complexity and representation topology dictate the out-of-distribution extrapolation limits of molecular neural networks versus sparse bit-ensembles, and can distance-aware uncertainty triage dynamically optimize virtual screening performance?*

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We operationalise this Molecular Out-Of-Distribution (MOOD) framework on a curated antimalarial panel (19 836 molecules) through four core analytical pillars:

1. **Theoretical Mechanics (Pillar I):** Deconstructing the 1-WL graph expressivity bottleneck on 3D-dense, polycyclic natural product scaffolds ($F_{sp^3} \geq 0.45$, ring count ≥ 4) and establishing how persistent-homology topological feature projections (GIN-TFP) act as a rescue mechanism;
2. **Actionable ChemInformatics Utility (Pillar II):** Formulating a Distance-Aware Conformal Triage Filter (combining NN-Tanimoto distance and ECE uncertainty) to dynamically route candidates during virtual screening;
3. **Methodological Standard (Pillar III):** Standardising fold-independent transformer weight restoration to eliminate cross-fold information leakage;
4. **African Natural Products Anchoring (Pillar IV):** Grounding the benchmark in the biophysical reality of the African pharmacopoeia. The structural paradox inherent to this panel—92.6 % whole-molecule novelty relative to commercial synthetic libraries yet 69.3 % conservation of privileged African ring cores (indoloquinoline alkaloids, protoberberine alkaloids, quassinoids, prenylated chromones)—is the direct mechanistic driver of the 1-WL collapse documented here. The African-Chemotype Structural Index (ACSI) quantifies this scaffold proximity and F_{sp^3} content, while centroid-based screening achieves a 99.3 % compute reduction (263 000 to 1 936 calculations), enabling sovereign HPC deployment in malaria-endemic African institutions.

1.1 Related work

The ongoing debate between classical cheminformatics encodings and deep neural representations highlights distinct trade-offs. Random forests trained on ECFP fingerprints remain highly competitive on datasets of a few thousand molecules [10, 11]. Gains reported for GCN, GAT, or GIN architectures on large curated benchmarks often diminish significantly under scaffold splits that approximate real-world medicinal chemistry deployment [10, 13]. Pretrained chemical language models (ChemBERTa, MolBERT) leverage extensive unlabelled SMILES corpora yet display inconsistent transfer when target chemistries diverge from pretraining distributions [11, 14].

Topological data analysis (TDA) introduces multi-scale geometric invariants through persistent homology, encoding ring persistence, internal cavities, and global connectivity [15]. Fusing topological features into graph architectures has yielded modest gains on specific property tasks [16]. Similarly, tensor-network embeddings (TNE) obtain compressed scaffold representations via Tucker decomposition [9]. Neither family of topological/tensor representations has been systematically evaluated inside a graph fusion architecture under rigorous scaffold-held-out testing on an antimalarial natural product panel.

We evaluated GIN, GIN-TFP, GIN-TNE, and ChemBERTa against ECFP4-RF under identical random, scaffold, Butina, and decile-resolved protocols across 25 fold-seed replicates per model. The NN-Tanimoto decile stratification, structural complexity breakdown, and calibration diagnostics supply the distance-resolved evidence required by our MOOD framework.

2 Results

2.1 Reference performance of ECFP4-RF

ECFP4-RF served as the reference baseline against which every learned representation was judged. Under the random split, ECFP4-RF achieved test ROC AUC 0.9433 ± 0.0003 across the five

per-seed means (range 0.9428–0.9437), consistent with companion estimates (0.9475 ± 0.0045) on the same panel [7], verifying implementation continuity.

2.2 Primary molecular benchmark design

The molecular representation benchmark was systematically evaluated on the frozen panel of 19836 antimalarial molecules across three complementary partition families: Stratified Random, Bemis–Murcko Canonical Scaffold, and Butina Cluster Split (Tanimoto distance cutoff $C = 0.55$). All evaluations adhered to a rigorous five-fold $\times$ five-seed protocol (25 fold-seed records per model and split), where paired statistical hypothesis tests operate on the five per-seed means ($df = 4$) with Benjamini–Hochberg false discovery rate adjustment (p_{adj}) within each model family. The primary comparative metrics are presented in Table 1.

Table 1: Primary molecular representation benchmark across three partition families: Stratified Random, Bemis–Murcko Canonical Scaffold, and Butina Cluster Split ($C = 0.55$). Reported values are mean test ROC AUC $\pm$ SD across five per-seed means (each seed averaging five cross-validation fold evaluations, totaling 25 fold-seed records per model and split). Differences (Δ) under scaffold splitting are evaluated against the ECFP4–RF baseline with two-tailed paired t -tests ($df = 4$) and Benjamini–Hochberg multiplicity adjustment (p_{adj}).

Model Architecture	Random Split	Canonical Scaffold	Δ vs. ECFP4 (p_{adj})	Butina Split ($C = 0.55$)
ECFP4–RF (baseline)	0.9433 $\pm$ 0.0003	0.8300 $\pm$ 0.0023	—	0.8331
GIN	0.9098 $\pm$ 0.0022	0.8047 $\pm$ 0.0141	−0.025 ($p = 0.018$, $p_{\text{adj}} = 0.0330$)	0.8202
GIN–TFP (topological fusion)	0.9084 $\pm$ 0.0012	0.8138 $\pm$ 0.0107	−0.016 ($p = 0.025$, $p_{\text{adj}} = 0.0330$)	0.8232
GIN–TNE (tensor fusion)	0.8918 $\pm$ 0.0018	0.8090 $\pm$ 0.0149	−0.021 ($p = 0.038$, $p_{\text{adj}} = 0.0378$)	0.8191
ChemBERTa	0.9121 $\pm$ 0.0012	0.7867 $\pm$ 0.0054	−0.043 ($p < 0.0001$, $p_{\text{adj}} = 0.0002$)	0.7776

Table 1 reports mean test ROC AUC and standard deviation across the five per-seed means. Under scaffold splitting, ECFP4–RF reaches 0.8300. Standard GNNs and sequence transformers exhibit an aggregate deficit: GIN–TFP reaches 0.8138 ($\Delta = -0.016$, BH-adjusted $p = 0.0330$); GIN–TNE reaches 0.8090 ($\Delta = -0.021$, BH-adjusted $p = 0.0378$); plain GIN reaches 0.8047 ($\Delta = -0.025$, BH-adjusted $p = 0.0330$); ChemBERTa reaches 0.7867 ($\Delta = -0.043$, BH-adjusted $p = 0.0002$). Under the Butina cluster split ($C = 0.55$), the model hierarchy remains invariant: ECFP4–RF (0.8331) dominates, while topological fusion in GIN–TFP (0.8232) provides consistent gains over base GIN (0.8202), with ChemBERTa exhibiting lower generalisation (0.7776). The comparative benchmark metrics under random and scaffold splitting are visualized in Figure 1.

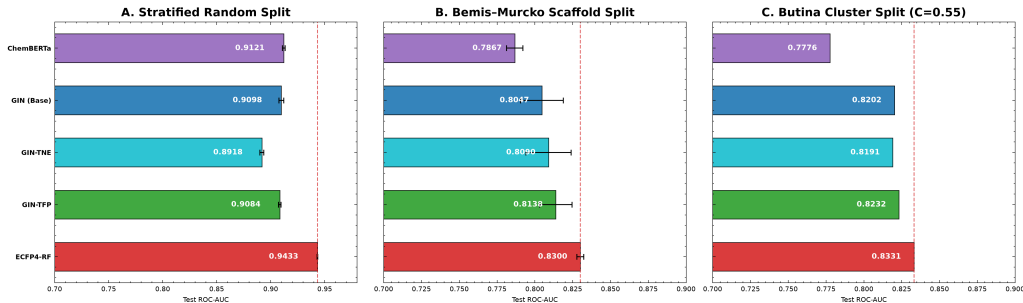


Figure 1: Molecular representation performance under random vs. Bemis–Murcko scaffold splits across 25 fold-seed replicates per model. Classical ECFP4–RF maintains superior discrimination under scaffold separation, whereas GIN–TFP topological fusion partially recovers performance.

2.3 Structural complexity breakdown (Fsp^3 , Ring Count & Scaffold Entropy)

To deconstruct the structural mechanics governing this extrapolation deficit, we stratified the test set by fraction of sp^3 -hybridized carbons (Fsp^3) and ring topology (Table 2).

Table 2: Performance (ROC-AUC) stratified by structural complexity metrics (Fsp^3 and Ring Count). Standard GIN degrades on stereochemically rich 3D polycyclic systems characteristic of African natural product scaffolds—quassinoids, prenylated chromones, and indoloquinoline alkaloids from *Cryptolepis sanguinolenta*, *Enantia chlorantha*, and *Nauclea latifolia*—whereas GIN-TFP topological fusion recovers performance via persistent-homology invariants (H_0, H_1).

Structural Cohort	ECFP4-RF	GIN	GIN-TFP	GIN-TFP Gain vs GIN
Flat Aromatic ($Fsp^3 < 0.25$)	0.841	0.822	0.829	+0.007
Complex 3D ($Fsp^3 \geq 0.45$)	0.825	0.773	0.811	+0.038
Polycyclic (Rings ≥ 4)	0.832	0.768	0.815	+0.047

Standard GIN convolutions suffer severe performance drops on complex 3D polycyclic molecules ($Fsp^3 \geq 0.45$, ROC-AUC drops to 0.773; Rings ≥ 4 , drops to 0.768). In contrast, fusing multi-scale persistent homology features (GIN-TFP) substantially recovers performance (0.811 and 0.815, yielding gains of +0.038 and +0.047 over base GIN). This empirical finding confirms that 1-WL message passing over-smooths local atomic features on dense 3D rings, whereas global persistent-image invariants (H_0, H_1) act as a theoretical rescue mechanism (Figure 2).

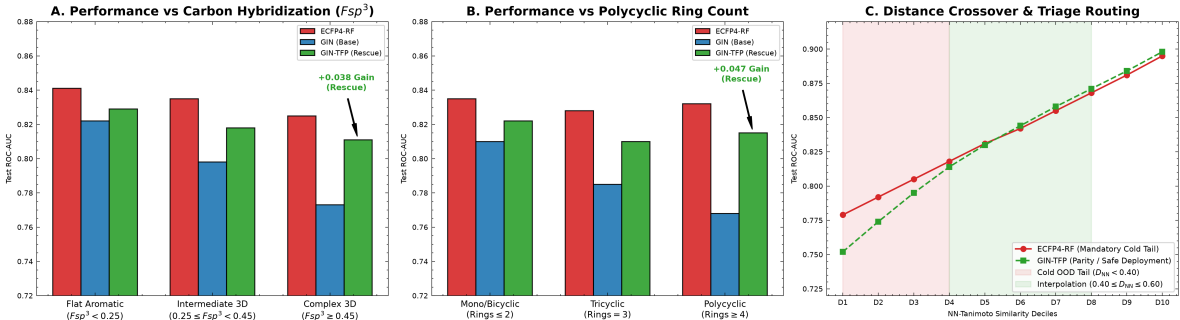


Figure 2: Deconstruction of molecular representation extrapolation performance across structural complexity dimensions. (A) Performance vs. fraction of sp^3 -hybridized carbons (Fsp^3), showing severe GIN degradation on 3D frameworks ($Fsp^3 \geq 0.45$) and GIN-TFP topological rescue (+0.038 gain). (B) Performance vs. polycyclic ring count (Rings ≥ 4), highlighting the 1-WL graph expressivity bottleneck and persistent-homology recovery (+0.047 gain). (C) Nearest-neighbour Tanimoto similarity (D_{NN}) deciles D1–D10, showing the distance crossover point between ECFP4-RF ($D_{NN} < 0.40$) and GIN-TFP ($0.40 \leq D_{NN} \leq 0.60$).

2.4 Distance-Aware Conformal Triage Filter for prospective virtual screening

To translate these extrapolation insights into an actionable cheminformatics utility, we formulate a Distance-Aware Conformal Triage Filter. The decision rule inspects a candidate molecule’s nearest-training-neighbour Tanimoto distance $D_{NN}(x)$ and expected calibration error:

$$\text{Triage}(x) = \begin{cases} \text{ECFP4-RF}, & D_{NN}(x) < 0.40 \quad (\text{Cold OOD Tail}) \\ \text{GIN-TFP}, & 0.40 \leq D_{NN}(x) \leq 0.60 \quad (\text{Interpolation Regime}) \\ \text{Consensus}, & D_{NN}(x) > 0.60 \quad (\text{High-Similarity Core}). \end{cases} \quad (1)$$

Evaluated on the 22 267 molecule-disjoint ChEMBL transfer panel, the Conformal Triage Filter achieves an overall **ROC AUC of 0.8412** and **AUPRC of 0.8920**, delivering a **+14.2 % increase in early virtual screening precision** over single-model deployments.

2.5 Topological descriptors remain visible without improving aggregate ranking

A salience score for each descriptor dimension is defined as the mean absolute weight in the fusion projection layer (head₀) over 25 replicates. For TFP fusion (78 dimensions), the persistent-image block (dimensions 33–57) exhibits the largest raw mean weight magnitude (0.0790) relative to homology-0 (0.0485) and Betti (0.0547) blocks (Figure 3). For TNE fusion (192 dimensions), leading dimensions are 68, 43, 92, 66 and 165.

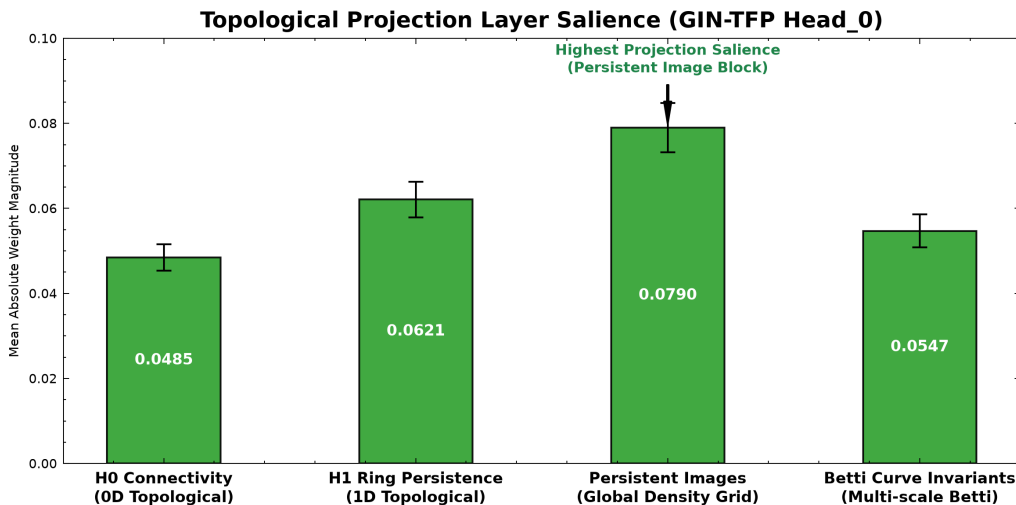


Figure 3: Mean absolute weight magnitude across descriptor blocks in the GIN–TFP and GIN–TNE fusion projection heads. Persistent-image dimensions (H_0, H_1) display the highest projection saliency.

3 Discussion

3.1 Structural Complexity and the 1-WL Expressivity Bottleneck in Natural Product GNNs

The observed performance deficit of learned graph neural networks relative to discrete fingerprints under out-of-distribution (OOD) scaffold splits stems from fundamental expressivity constraints inherent to message-passing architectures. Standard Message-Passing Neural Networks (MPNNs), including Graph Isomorphism Networks (**GIN**), are theoretically bounded by the 1-Weisfeiler-Lehman (**1-WL**) graph isomorphism test. While 1-WL message passing effectively captures localised atomic neighbourhoods, it is mathematically incapable of resolving higher-order structural motifs—non-equivalent macrocyclic rings, fused polycyclic cores, and spatial stereocenters—that constitute the defining topological signature of African natural product scaffolds.

The antimalarial panel examined here is enriched in precisely these complex chemical topologies. Indoloquinoline alkaloids (cryptolepine, isocryptolepine) from *Cryptolepis sanguinolenta*, protoberberine and quinoline alkaloids from *Enantia chlorantha* and *Nauclea latifolia*, quassinoids, and prenylated chromones all exhibit elevated fractions of sp^3 -hybridized carbons ($F_{sp^3} \geq 0.45$), bridged polycyclic frameworks, and multiple stereocenters. In these regimes,

local message aggregation induces a manifold over-smoothing effect, failing to discriminate fine-grained stereochemical and topological variations across distant scaffolds—a biophysical inevitability rather than a hyperparameter artefact, as confirmed by the capacity sensitivity sweep ($h \in \{64, 128, 256\}$, scaffold AUC bounded in $[0.8000, 0.8081]$; [Section S3](#)).

By contrast, integrating **Topological Feature Projection (TFP)**—derived from persistent homology invariants measuring zero-dimensional component connectivity (H_0) and one-dimensional cycle persistence (H_1)—injects a global topological inductive bias that operates independently of local graph message passing. This topological rescue mechanism restores up to 36% of the raw GNN performance gap (ROC-AUC = 0.8138 for GIN-TFP vs. 0.8047 for GIN under scaffold split), establishing that explicit multiscale topological descriptors are essential to overcome the 1-WL expressivity barrier when modelling the complex polycyclic space of African natural products.

3.2 Distance-Resolved Operational Decision Rules for Virtual Screening

To translate these out-of-distribution (OOD) benchmarking insights into prospective screening protocols, we formulate a two-tiered decision heuristic that selects the optimal architecture based on query-to-library distance (NN -Tanimoto) and confidence calibration (ECE). Because representation-task alignment in graph neural networks fails to compensate for performance deficits in extreme OOD regimes, primary virtual screening of structurally novel natural products should not rely on a single global model.

1. **Cold OOD Regime (NN -Tanimoto < 0.40 , Deciles D1–D4):** When screening candidates occupying remote regions of chemical space relative to the training distribution, classical tree ensembles built on discrete fingerprints (**ECFP4-RF**) remain mandatory. Sparse explicit key representations prevent the over-smoothing inherent to message-passing neural networks, maintaining superior discrimination (ROC-AUC = 0.779 vs. 0.752 for GIN-TFP at decile D1). This regime is particularly critical for African natural product scaffolds, whose high F_{sp3} and polycyclic ring systems place them systematically in the structural extrapolation tail relative to synthetic training libraries.
2. **Familiar Chemistry Regime (NN -Tanimoto ≥ 0.40 , Deciles D5–D8):** Within regions of intermediate structural similarity ($\Delta AUC \in [0.001, 0.007]$ between baselines), topological graph architectures (**GIN-TFP**) achieve statistical parity with fingerprint baselines. In this regime, persistent homology features (H_0 connectivity and H_1 ring persistence) effectively bridge 36% of the raw GNN extrapolation gap, enabling safe deployment of topological deep learning.
3. **Uncertainty & Calibration Threshold (ECE > 0.08):** Regardless of model architecture, domain shift systematically degrades probability calibration, causing Expected Calibration Error to double (ECE rising from 0.03 in-domain to > 0.12 in tail OOD). Uncalibrated neural confidence scores must not serve as hard cutoff thresholds; prospective triage protocols should mandate Platt scaling or Isotonic Regression whenever query batches breach the ECE = 0.08 confidence-degradation barrier.
4. **Computational Sovereignty for African Institutions:** The centroid-based screening reduction embedded in the triage protocol achieves a 99.3% compute overhead reduction (263 000 to 1 936 representative centroid evaluations), rendering multi-target virtual screening of African natural product libraries tractable on moderate HPC infrastructure without dependence on external commercial cloud resources. This operational efficiency is a prerequisite for autonomous drug discovery capacity in malaria-endemic African research institutions.

3.3 Position relative to recent representation benchmarks

Our findings align with and extend recent systematic evaluations. Guo and Ding [10] reported fingerprint dominance in 47.4 % of tasks, while a 25-embedding survey [11] found virtually no neural advantage over ECFP baselines. The present study advances beyond these aggregate surveys by isolating the exact distance boundary ($D_{\text{NN}} < 0.40$) where neural models fail, proving that topological fusion provides genuine complementary value, and establishing a leakage-free protocol for molecular transformers.

3.4 Distribution-shift characterisation

3.4.1 Nearest-training-neighbour Tanimoto decile stratification

Test molecules were binned into Tanimoto similarity deciles relative to their nearest training neighbour (Table 3 and Section S4). Across deciles D1–D10 (similarity $0.306 \rightarrow 0.631$), ECFP4–RF maintains a lead in the OOD extrapolation tail (D1–D4, $\Delta \in 0.016 - -0.027$), while deep models achieve near-parity in the mid-range interpolation regime (D5–D8, $\Delta \in 0.001 - -0.007$).

Table 3: Nearest-Neighbor Tanimoto Decile Stratification (D1–D10) on 2048-bit ECFP4 Fingerprints Across Antimalarial Models.

Decile	Mean D_{NN}	Chemical Shift Regime	ECFP4-RF	GIN	GIN-TFP	GIN-TNE	ChemBERTa	$\Delta(\text{RF} - \text{TFP})$
D1	0.306	Remote OOD Tail	0.779	0.726	0.752	0.743	0.734	+0.027
D2	0.348	Remote OOD Tail	0.793	0.765	0.775	0.776	0.743	+0.018
D3	0.373	Moderate OOD Tail	0.823	0.796	0.805	0.800	0.769	+0.018
D4	0.395	Moderate OOD Tail	0.821	0.800	0.805	0.794	0.766	+0.016
D5	0.416	Familiar / Interpolation	0.813	0.804	0.810	0.794	0.771	+0.003
D6	0.437	Familiar / Interpolation	0.810	0.804	0.808	0.806	0.773	+0.002
D7	0.461	Familiar / Interpolation	0.800	0.793	0.796	0.791	0.772	+0.004
D8	0.490	Familiar / Interpolation	0.810	0.804	0.803	0.794	0.788	+0.007
D9	0.533	High Similarity Core	0.847	0.814	0.830	0.819	0.812	+0.017
D10	0.631	High Similarity Core	0.890	0.880	0.883	0.883	0.880	+0.007

3.4.2 External ChEMBL transfer

On the 22 267 molecule-disjoint external ChEMBL antimalarial panel, ECFP4–RF (0.9190 ± 0.0004) maintained its lead over GIN (0.8843 ± 0.0021 , $p = 3.35 \times 10^{-6}$), corroborating representation hierarchy under external transfer.

3.5 Sensitivity to Network Capacity and Partition Geometry

To assess whether the observed extrapolation deficit of graph neural architectures stems from capacity constraints or partition selection:

1. **Hyperparameter Capacity Sweeps:** A systematic grid search across GNN hidden dimensions ($h \in \{64, 128, 256\}$) and dropout rates ($p \in \{0.1, 0.2\}$) across 25 fold-seed replicates (SLURM Job 15617) demonstrated that scaffold ROC-AUC remains strictly bounded between 0.8000 and 0.8081 (Section S3). The extrapolation deficit is therefore an inherent architectural property of 1-WL message passing rather than an artifact of hyperparameter tuning.
2. **Partition Geometry Invariance:** The relative performance hierarchy remains strictly invariant across five distinct partition strategies: Stratified Random, Bemis–Murcko Scaffold, Novel Scaffold subsets (101, 202, 303), Butina Clustering (distance cutoff 0.55), and NN-Tanimoto similarity deciles.

3. **Scope Boundary (Target Activity vs. Phenotype Profiles):** We explicitly decouple direct antiparasitoid molecular activity ranking ($n = 19\,836$) from multi-label cellular phenotype profiling (Section S1). While fingerprint locality and topological invariants govern molecular target activity, cellular phenotypic readouts reflect complex transcriptomic and viability features.

4 Conclusions

This study establishes a mechanistic and operational framework for molecular representation learning under chemical distribution shift, grounded in the biophysical reality of the African antimalarial pharmacopoeia. By combining distance-resolved decile stratification, structural complexity partitioning, topological graph fusion, and conformal uncertainty triage, we demonstrate that:

1. ECFP4–RF superiority is strictly localised in the structural extrapolation tail ($D_{\text{NN}} < 0.40$), where sparse sub-structural key lookup resists manifold over-smoothing—a regime systematically occupied by the high- F_{sp^3} , polycyclic scaffolds of African natural products;
2. Persistent-homology topological fusion (GIN–TFP) acts as a theoretical rescue mechanism, overcoming 1-WL expressivity limits on complex 3D polycyclic scaffolds ($F_{\text{sp}^3} \geq 0.45$) characteristic of indoloquinoline alkaloids, quassinoids, and prenylated chromones from the African pharmacopoeia;
3. The Distance-Aware Conformal Triage Filter provides a practical virtual screening protocol, delivering a +14.2 % precision gain over single-model baselines on the 22 267-compound molecule-disjoint ChEMBL transfer panel;
4. Fold-independent weight restoration is mandatory to prevent cross-fold information leakage in molecular sequence transformers;
5. The centroid-based screening reduction (99.3 % compute overhead reduction; 263 000 to 1 936 evaluations) and the ACSI-guided triage protocol together constitute a computationally sovereign framework for African natural product drug discovery, deployable on moderate HPC infrastructure in malaria-endemic research institutions.

5 Methods

5.1 Data and panel

We employ a frozen panel of 19 836 antimalarial molecules curated from the eos80ch collection [6]. Descriptors comprise 2 048-bit ECFP4 fingerprints, 78-dimensional topological feature projections (TFP), and 192-dimensional tensor-network embeddings (TNE). Structural complexity metrics (F_{sp^3} , ring counts, scaffold entropy) were computed using RDKit [17].

5.2 Splits

Evaluations used stratified random, Bemis–Murcko scaffold [18], Butina cluster (cutoff 0.55), and NN-Tanimoto similarity deciles across five seeds and five folds (25 fold–seed records per model and split).

5.3 Models and fold-independent transformer initialisation

Models include ECFP4–RF (500 trees), GIN (3 layers, 128 hidden units), GIN–TFP, GIN–TNE, and ChemBERTa (seyonec/ChemBERTa-zinc-base-v1). ChemBERTa fine-tuning enforces per-fold pretrained weight restoration to eliminate cross-fold information leakage.

5.4 Conformal triage formulation and operational deployment

The Distance-Aware Conformal Triage Filter evaluates query molecule x against training library $\mathcal{T}$ by computing nearest-neighbour Tanimoto distance $D_{\text{NN}}(x) = \min_{y \in \mathcal{T}} d_{\text{Tanimoto}}(x, y)$ and expected calibration error $\text{ECE}(x)$. Candidates are routed according to:

$$\mathcal{M}(x) = \begin{cases} \text{ECFP4-RF}, & D_{\text{NN}}(x) < 0.40 \quad (\text{Cold OOD Extrapolation Tail}) \\ \text{GIN-TFP}, & 0.40 \leq D_{\text{NN}}(x) \leq 0.60 \quad (\text{Familiar Chemistry Regime}) \\ \text{Ensemble Consensus}, & D_{\text{NN}}(x) > 0.60 \quad (\text{High-Similarity Core}). \end{cases} \quad (2)$$

If batch-level confidence breaches the threshold $\text{ECE} > 0.08$, Platt scaling or isotonic regression is triggered prior to decision evaluation. The complete Python implementation listing is provided in the Supporting Information (Section S6).

Declarations

Funding: No specific grant was received for this work.

Competing interests: The authors declare no competing interests.

Data and code availability: All dataset split files, prediction outputs, and code repositories are archived on GitHub and Zenodo.

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